

6

(b) Describe what is meant by robotics.

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..... [2]

(d) The robot needs to find its way through different puzzles. Each puzzle has a series of paths that the robot needs to follow to find its way to the end of the puzzle. The puzzle contains dead ends and obstacles, so the robot needs to decide which way to go.

The robot's program will use artificial intelligence (AI).

(i) Describe the characteristics of AI.

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..... [3]

(ii) Explain how the program will use AI.

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..... [4]

9 A company uses robots in its factory to manufacture large pieces of furniture.

(a) One characteristic of a robot is that it is programmable.

State **two** other characteristics of a robot.

1

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2

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[2]

(b) Give **two** advantages to company employees of using robots to manufacture large pieces of furniture.

1

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2

.....

[2]

(c) Give **one** disadvantage to the company's owners of using robots to manufacture large pieces of furniture.

.....

..... [1]

9 One component of an expert system is the inference engine.

(a) Identify the **three** other components in an expert system.

1

2

3 [3]

(b) Describe the role of the inference engine in an expert system.

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..... [2]

7 The rule base and the inference engine are two components of an expert system.

(a) Identify the other **two** components of an expert system.

1

2 [2]

(b) Describe the role of the rule base in an expert system.

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..... [2]

(b) Explain **one** advantage to employees of using an automated system in manufacturing.

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..... [2]

(c) Explain **one** disadvantage to the company owner of using an automated system in manufacturing.

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..... [2]

10 A car repair garage uses an expert system.

(a) Complete the description about the operation of the expert system.

Use the terms from the list. Some of the terms in the list will **not** be used.

inference engine interface knowledge base
machine learning mechanical engine output device
question base rule base

An expert system has a that contains a list of facts.

The applies the

to the to reach a diagnosis for the repair of the car.

The user provides data to the system using an

[5]

(b) The expert system has machine learning capabilities.

Describe what is meant by machine learning capabilities.

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..... [4]